

451/1

COMPUTER STUDIES

PAPER 1

(THEORY)

TIME: 2 ½ hrs



ROCKFIELDS SENIOR SCHOOL

Soaring To Greatness

NAME:.....ADMNO.:.....CLASS.....

FORM 3

MID -TERM 1 EXAMINATION 2026

COMPUTER STUDIES

INSTRUCTION TO CANDIDATES

- Write your name and admission number in the spaces provided above.
- This paper consists of **Two** sections **A** and **B**.
- Answer **ALL** questions in section **A**.
- Answer question 16 and any three in section B.
- All answers should be written in the spaces provided on the question paper.

FOR EXAMINER'S USE ONLY

SECTION	QUESTIONS	CANDIDATE'S SCORE
A	1 -15	
B	16	
	17	
	18	
	19	
	20	
	TOTAL SCORE	

This paper consists of 9 Printed pages

Student should check the question paper to ensure that all the pages are printed as indicated and no question are missing.

SECTION A (40MARKS)

Answer all questions in this section

1. Give an example of the following computer components (2marks)
 - i) Input devices
.....
 - ii) Output devices
.....
2. Differentiate between Central processing unit and system unit: (2marks)
.....
.....
3. State three characteristics of the first generation of computers. (3marks)
.....
.....
.....
4. Give three disadvantages of using computer today. (3marks)
.....
.....
.....
5. Before setting up a computer lab Shujaa School was advised to make several considerations to ensure they put up a favorable lab.
 - a) What are the factors to be considered when preparing a computer lab? (3marks)
.....
.....
.....
 - b) What measures should Shujaa school put in place to ensure the security of:
 - i) The computers in the lab. (2marks)
.....
.....
 - ii) The users of the lab. (2marks)
.....
.....
6. Describe each of the following computer cable. (2marks)
 - i) Parallel cable
.....
 - ii) Serial cable
.....

7. Write the following acronyms in full (2marks)
- i) RAM_____
- ii) CD-ROM_____
8. Outline two advantages of hard disk over floppy disk. (2marks)
-
-
9. State two reasons to justify why computer users are always advised to follow the right Procedure of shutting down (2marks)
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-
10. Mention three reasons why a calculator is a computer. (3 marks)
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-
-
11. Give two causes of computer data loss. (2 marks)
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-
12. List three factors that determine the cost of a computer. (3 marks)
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-
-
13. State three characteristics which make computers better than human beings. (3 marks)
-
-
-
14. Explain the term portability in reference to hardware and software. (2marks)
-
-
15. Name two types of optical disks. (2marks)
-
-

SECTION B (60 MKS)

Answer question 16 and any other three questions from this section in the spaces provided

16. a) State **three** factors considered when choosing an operating system for use in a computer. (3 marks)

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- b) An operating system manages and organizes a computer system using the following structures:

File, Folder and **Drive**. State the meaning of the underlined items respectively.

(3 marks)

.....

.....

.....

- c) Differentiate between error handling and job scheduling as used in operating systems. (2marks)

.....

.....

.....

- d) Disk management is one important aspect in secondary storage in computer systems. Explain the following tools used by an operating system to manage disks in the computer. (3 marks)

(i) Formatting

.....

.....

(ii) Partitioning

.....

.....

(iii) Defragmentation

.....

.....

(e) (i) Failure of information systems is a major concern in the security of data in computing systems. State **two** causes of such failure. (2 marks)

.....

.....

(ii) Explain the following computer crimes.

a) Fraud (1 mark)

.....

.....

b) Industrial espionage (1 mark)

.....

.....

17. a) State the number systems represented by the following numbers (3marks)

i) 5A9H_____

ii) 100101001₂_____

iii) 8903₁₀_____

b) Convert the following numbers to what is given in brackets.

i) 45₁₀(to binary) (3marks)

.....

.....

.....

.....

.....

ii) 4.75₁₀(to binary) (3marks)

.....

.....

.....

.....

.....

iii) 10.11₂ (to decimal) (3marks)

.....

.....

.....

.....

c) Explain the meaning of the following terms in reference to computer memory and data size. (3marks)

i) Byte

.....

ii) Bit

.....

iii) Nibble

.....

18. (a) Define a database model (2marks)

.....

.....

(b) List three advantages of using electronic database systems (3marks)

.....

.....

.....

(c) In a database system, data integrity ensures the correctness and completeness of the data in the database. Differentiate the following types of integrity constraints:

i. Validity integrity (1mark)

.....

.....

ii. Referential integrity (1mark)

.....

.....

(d) Briefly explain any two database models (4marks)

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.....

.....

.....

(e) A student' database comprises of students' details table and fees received table as shown below:

Students' details table

Surname
Middle Name
First Name
Admission Number
Course

Fees Received table

Date
Amount
Receipt No.
Admission Number

i) State the primary key field for each table. (2mks)

.....

ii) Which field is appropriate to use to create relationship between the two tables (1mark)

.....

iii) Name the foreign key in the Fees Received table (1mark)

.....

19. i) Briefly explain the following terms citing an appropriate example: (6marks)

a) Cell

.....

b) Range

.....

c) Label

.....

ii) Use the sample worksheet provided to answer the questions below.

	A	B	C	D	E	F
1	Items	Jan	Feb	Mar	April	Total
2	Fees	400	460	400	400	
3	Food	300	350	305	270	
4	Electricity	100	100	100	100	
5	Fuel	150	150	150	150	
6	Transport	380	270	150	300	
7	Total					

a) Write down the **data type** that is stored in the following cells.

(2marks)

i) C5_____

ii) A4_____

b) What cell contains the data item representing transport for the month of February?

(1mark)

.....

c) Write down the formula that may be used to compute;

i) The total in cell F6.

(1mark)

.....

ii) The lowest expenditure in cell D7

(1mark)

.....

d) Name any **two** types of **graphs** that may be supported by spreadsheets.

(1mark)

.....

.....

e) A formula E7*F11 was typed in cell G9 and later copied to cell C4. What is the formula in the

destination cell

(3marks)

.....

.....

.....

20. a) Describe each of the following computer crimes. (3marks)

i) Hacking

.....
.....

ii) Piracy

.....
.....

iii) Sabotage

.....
.....

b) Mwende noticed that her PC is taking too long to load programs and files. The computer technician informed her that there was a high possibility it a virus.

i) What is a computer virus? (1mark)

.....

ii) List any **three** measures that Mwende needs to put into place to protect her computer from virus infection. (3marks)

.....
.....
.....

d) State **three** advantages of word processors over manual typewriter. (3marks)

.....
.....
.....

e) List **three** editing features of word processing. (3marks)

.....
.....

f) Give two attributes of a file (2marks)

.....
.....